Will Cygan

Senior Software Engineer, Linkedin Business Platform

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Work Experience

Senior Software Engineer — Chicago, IL

LinkedIn — Mar 2024 – Present

- Architected real-time alerting system processing 100,000+ QPS using Kafka/Flink/Venice, recovering $2M+ annually in involuntary churn by notifying subscribers of failed payments before subscription cancellations.
- Eliminated N+1 query problem in Order Processing system, reducing average query latency by 40% (50ms⟶30ms) and p95 by 37% (200ms⟶125ms) across all ordering workflow read paths.
- Built reusable Oracle-to-MySQL migration framework adopted by 12 teams, saving 12+ months of cumulative engineering time and standardizing migration patterns across the organization.
- Executed zero-downtime Oracle-to-MySQL migration for 2 databases handling 3,000 QPS, maintaining strong consistency through Couchbase-backed sticky sessions and GoldenGate replication.
- Created Airflow cache invalidation job purging 30,000 stale records daily, preventing unnecessary gRPC calls to downstream services and reducing cross-team service load.
- Reduced Order database size by 33% (12TB⟶9TB) by implementing distributed deletion pipeline using Spark, Kafka, and batch processing to safely remove 3TB of obsolete records.
- Optimized JVM performance across 5 production services, improving health scores from 30-80% to 99.9%+ by reducing GC pause times by 86% (700ms⟶100ms) and eliminating daily alerts.

Software Engineer — San Francisco, CA

LinkedIn — Feb 2022 – Mar 2024

- Scaled Videos You Might Be Interested In recommendation to 3,000 QPS on LinkedIn Feed, increasing course discovery CTR by 10% through TikTok-style video carousel.
- Built Learning Alerts Spark pipeline analyzing 50TB+ weekly to match 10M+ job seekers with learning courses aligned to their career goals, increasing notification conversion rates by 5%.
- Developed deterministic bucketing algorithm for Learning Alerts Spark pipeline, enabling clean A/B test readouts by segmenting 10M+ users into isolated experiment groups.

Projects

Anton (Kubernetes Homelab) — github.com/wcygan/anton

- Built fault-tolerant 3-node bare-metal Kubernetes cluster using Talos Linux with GitOps CI/CD automation, achieving 99.9% uptime over 12+ months.
- Built and deployed e-commerce site (kneadybynaturebakery.com) on homelab cluster so my sister can sell online
- Deployed distributed data systems (TiDB, RedPanda, DragonflyDB, ScyllaDB, ClickHouse) to evaluate performance characteristics and operational trade-offs of modern database alternatives.

tokio-utils Rust Library — github.com/wcygan/tokio-utils

- Published async Rust library implementing rate limiting, object pooling, and graceful shutdown patterns, achieving 2x performance improvement in object pool benchmarks.
- Implemented web crawler processing 300+ pages/minute with adaptive rate limiting, achieving 0% IP blacklist rate across 1,000+ domains while building internet graph index for PageRank analysis.

Skills

Languages	Java, Rust, Python, SQL, Bash, Go, TypeScript, Scala
Technologies	Kafka, gRPC, Temporal, Flink, Spark, Airflow, Trino, Kubernetes, Docker
Databases	MySQL, Redis, Couchbase, Oracle, DragonflyDB, Venice, HDFS, S3

Education

University of Illinois at Chicago — Chicago, IL

B.S. in Computer Science — 2021